GREEN VISION
CLOUD

Unified Multi-Cloud Carbon Management.

MERN Stack powered platform

Real-time carbon emission tracking

Supports **AWS, Azure, and Google Cloud**

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The Carbon Visibility Challenge

Invisible Emissions

Fragmented cloud data hides true environmental impact.

Delayed Insights

Slow reports prevent real-time optimization and intervention.

Traditional carbon offsets are often unauditable and fail to align with strategic sustainability goals.

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The Imperative of Carbon Tracking

Managing carbon emissions is a strategic necessity for business resilience and growth.

Regulatory Compliance

Meet global regulations (CSRD, SEC) to avoid penalties and maintain market access.

Enhanced Brand Reputation

Attract conscious investors, customers, and top talent with strong environmental responsibility.

Operational Efficiency & Cost Savings

Optimise resource use and reduce energy consumption for tangible cost reductions.

Future-Proofing Business

Mitigate climate risks, foster innovation, and gain a competitive edge in a green economy.

Accurate carbon tracking is essential for informed decision-making and sustainable practices.

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Market Drivers & Regulatory Landscape

ESG Mandates

Investors demand transparent environmental reporting aligned with GHG Protocol.

FinOps + GreenOps

Enterprises integrate cost optimization with carbon reduction as core KPIs.

1 2 3

CSRD Compliance

CSRD requires detailed carbon disclosures for Scope 2 & 3 emissions.

Cloud providers expose only partial carbon data, creating normalisation challenges. Organisations require vendor-agnostic platforms to aggregate, compare, and act on emissions across their entire multi-cloud infrastructure.

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Our Three-Tier Solution

Unify & Quantify

Transform raw cloud usage into energy consumption and CO_2e emissions using precise regional carbon intensity factors across all major providers.

Live Tracking

Windowed aggregation delivers real-time trends, historical comparisons, and instant CSV exports for compliance and internal reporting.

Action Layer

Receive optimisation recommendations, plan offset strategies, and track coverage to ensure transparency and accountability across your sustainability programme.

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Cloud Data Pipeline Architecture

Ingest

Connect to AWS CloudWatch Logs and Azure Monitor APIs on configurable intervals. GCP integration ready for deployment.

Process

Transform raw bytes and compute minutes into kilowatt-hours, then into $kgCO_2$ using regional factors. Aggregate by time window and provider in MongoDB.

Serve

REST APIs return emission summaries, time-bucketed series, and provider comparisons for downstream consumption and analysis.

Present

React UI displays live KPIs, provider chips, sparklines, and stacked emission views with intuitive drill-down capabilities.

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Platform Objectives

01

Measure with Precision

Deliver accurate, real-time emissions tracking broken down by cloud provider, region, and service category.

02

Analyse for Efficiency

Surface trends, efficiency ratios (CO_2 per dollar, CO_2 per gigabyte), and outliers indicating waste or opportunities.

03

Act on Insights

Reduce emissions through optimisation recommendations; offset residual carbon with full audit trail and coverage tracking.

04

Report with Confidence

Enable simple CSV exports and API access for compliance reporting, ESG disclosures, and internal stakeholder communication.

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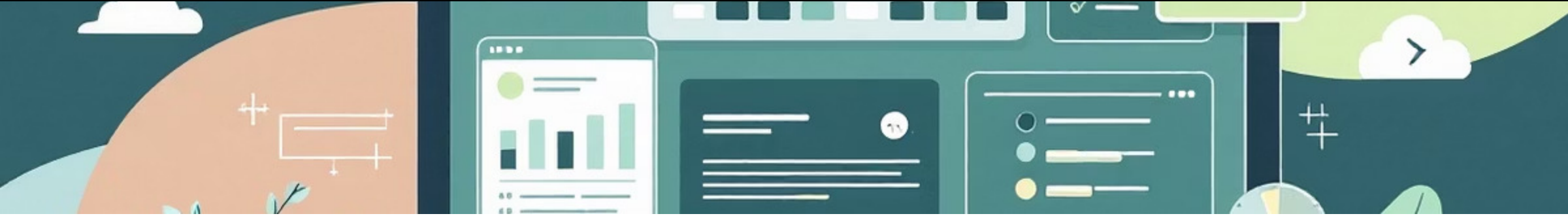

Products & Services Overview

1

Carbon Tracking Suite

Real-time emissions dashboards with provider breakdowns, regional intensity mapping, and time-series trend analysis.

2

Cost & Efficiency KPIs

Track CO_2 per dollar spent and CO_2 per gigabyte processed to identify wasteful workloads and optimisation opportunities.

3

Offsets Planner

Strategic carbon offset management with coverage tracking, audit trails, and integration-ready architecture for carbon registries.

Professional Services

- Multi-cloud setup and configuration consulting
- Optimisation advisory for workload placement and timing
- Compliance reporting and documentation support

API Access: `/api/cloud/carbon`, `/api/cloud/dashboard`, /api/ai/recommendations

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Strategic Position: SWOT Analysis

Strengths

- Real-time multi-cloud visibility
- Actionable, data-driven insights
- Powerful normalisation engine

Weaknesses

- Reliance on API stability
- Limited direct energy data
- GCP integration in progress

Opportunities

- Expand GCP full integration
- Registry-backed offset verification
- Strategic enterprise partnerships

Threats

- Competitor native carbon tools
- Evolving emission factors
- Market consolidation risk

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PESTEL Analysis

Political

Carbon pricing & regulatory incentives

Economic

Cost-carbon tradeoff optimisation

Social

ESG expectations from stakeholders

Technological

Greener instance types & adoption

Environmental

Grid mix variations by region

Legal

Mandatory carbon disclosure frameworks

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Scientific & Industry Foundations

GHG Protocol

Industry-standard framework defining Scopes 1, 2, and 3, plus reporting best practices for organisational carbon accounting.

Estimation Chain

Cloud usage (compute, storage, network) → kilowatt-hours consumed → $kgCO_2$ via regional grid carbon intensity factors.

Reduction–First Principle

Prioritise operational efficiency and workload optimisation before applying high-quality offsets with verified additionality and permanence.

> "Accurate measurement is the foundation of effective reduction. Without real-time visibility, sustainability goals remain aspirational rather than operational."

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Research Methodology

Design Approach

Observational methodology with time-windowed metrics to capture usage patterns and emission trends.

Quantification Model

Factor-based conversion leveraging regional carbon intensity and provider-specific PUE estimates.

Validation Protocol

Calculations cross-referenced against provider disclosures and workload benchmarks for accuracy.

Evaluation Framework

Track pre/post-optimization emissions, offset coverage, and cost-neutrality of interventions.

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Driving Sustainable Cloud Futures

Unlock Carbon Transparency

Gain deep, real-time visibility across all your multi-cloud providers, transforming hidden emissions into clear, actionable data.

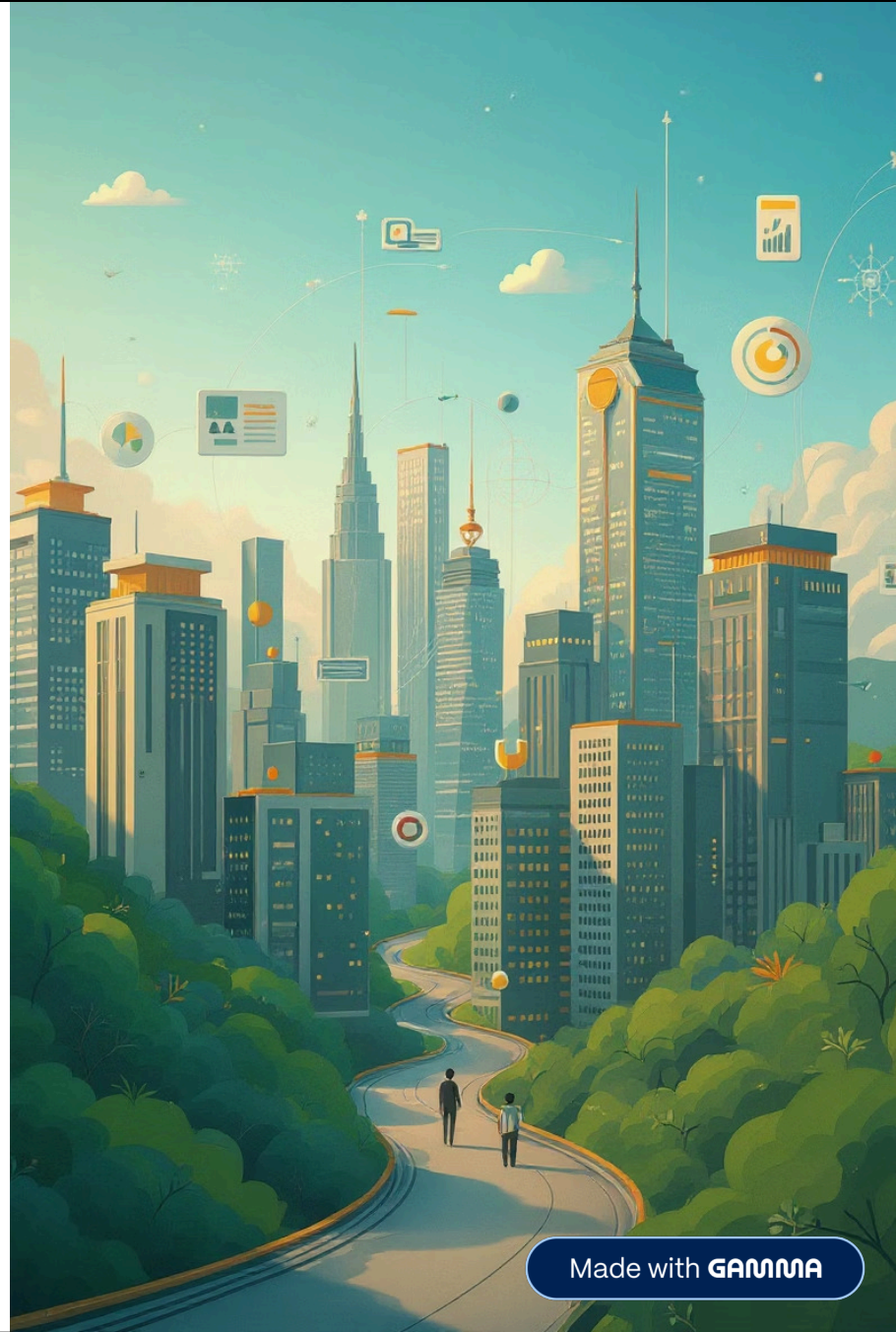
Drive Actionable Reduction

Leverage data-driven insights and optimisation recommendations to achieve significant emission cuts and improve cost efficiency.

Ensure Compliance & Growth

Meet global ESG mandates and CSRD requirements with auditable reporting, fostering a future-proof, sustainable cloud strategy.

GreenVision Cloud empowers your organisation to measure, analyse, act, and report on cloud carbon with unparalleled precision and confidence.



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Thank You!

We appreciate you exploring GreenVision Cloud. Together, let's empower your organisation's sustainability journey with clear, confident carbon tracking and contribute to a greener future.

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